

Triagegeist

AI-Based Emergency Triage Severity Prediction

Project Report

Machine Learning for Emergency Department Decision Support

Group Members

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1. Executive Summary

Triagegeist is a machine-learning-powered clinical decision support system designed to predict patient acuity levels in emergency departments. By analysing structured intake data — vital signs, chief complaint, demographics, and arrival mode — the system classifies each patient into one of three severity tiers (Low, Medium, High), enabling faster and more consistent triage decisions.

This report documents the full development lifecycle: problem framing, exploratory data analysis, preprocessing and feature engineering, comparative model evaluation across six algorithms, hyperparameter optimisation, and clinical interpretation of results.

2. Introduction & Problem Statement

2.1 Background

Emergency departments worldwide operate under constant pressure. Patient volumes frequently exceed available resources, forcing triage nurses to make rapid severity assessments with incomplete information and significant cognitive load. Studies have shown inter-rater variability in triage scoring can reach 30–40%, contributing to both over-triage (wasted critical resources) and under-triage (delayed treatment for high-acuity patients).

Artificial intelligence offers a complementary layer of decision support — not to replace clinical judgement, but to provide a data-driven second opinion, particularly in high-volume or under-staffed settings.

2.2 Consequences of Triage Errors

- Under-triage: life-threatening conditions go undetected, increasing patient mortality.
- Over-triage: low-severity patients consume critical-care resources, degrading overall throughput.
- Inconsistency: different nurses may assign different acuity scores to identical presentations.
- Burnout: overburdened staff make more errors during peak hours.

2.3 Objectives

- Train a multi-class classification model to predict triage severity (Low / Medium / High).
- Compare six diverse ML algorithms on identical data to identify the strongest performer.
- Engineer clinically meaningful composite features to improve predictive power.
- Provide interpretable feature-importance analysis to support clinical trust.
- Deploy an interactive web application for real-time inference.

3. Dataset Description

3.1 Source & Structure

The dataset originates from the Kaggle Emergency Triage competition. The notebook includes a synthetic data generator — producing 5,000 training and 1,000 test records — when competition files are unavailable, ensuring end-to-end reproducibility without a Kaggle account.

3.2 Features

Feature	Type	Clinical Significance
Age	Numerical	Elderly (>75) and paediatric (<5) patients carry elevated risk weights.
Heart Rate	Numerical	Tachycardia (>130 bpm) or bradycardia (<50 bpm) flags cardiovascular instability.
Systolic BP	Numerical	Hypotension (<90 mmHg) is a key shock indicator; hypertension (>180) suggests crisis.
Diastolic BP	Numerical	Used alongside systolic to compute pulse pressure (engineered feature).
Temperature	Numerical	Fever (>38°C) or hypothermia (<36°C) signals infection or systemic compromise.
SpO ₂	Numerical	Oxygen saturation below 94% indicates hypoxia requiring urgent intervention.
Resp. Rate	Numerical	Tachypnoea (>30) or bradypnoea (<10) reflect respiratory compromise.
Pain Scale	Ordinal 0–10	Patient-reported pain; ≥8 contributes to high-severity classification.
Chief Complaint	Categorical	Free-form complaint encoded; chest pain and altered consciousness are high-risk.
Gender	Categorical	Demographic factor; lower direct weight in tree models.
Arrival Mode	Categorical	Ambulance arrivals receive a severity bonus reflecting pre-hospital assessment.

3.3 Target Variable

The severity column contains three ordinal classes: Low, Medium, and High. Under synthetic generation, class proportions are approximately 55% Low, 30% Medium, and 15% High — reflecting realistic emergency department distributions where the majority of walk-in patients present with non-critical conditions.

3.4 Data Quality

Approximately 8% of values in five continuous vitals columns (heart_rate, systolic_bp, temperature, spo2, resp_rate) were intentionally nullified during synthetic generation to simulate real-world data incompleteness. All missing values were imputed using column-wise medians prior to modelling.

4. Methodology

4.1 Exploratory Data Analysis

EDA proceeded through three lenses. First, target distribution was examined — bar and pie charts confirmed mild class imbalance (Low > Medium > High). Second, feature distributions were visualised with overlapping histograms stratified by severity class, revealing clear separation in SpO₂, heart rate, and blood pressure between High and Low severity groups. Third, a Pearson correlation heatmap identified modest positive correlation between systolic and diastolic BP, and between age and pain scale.

4.2 Preprocessing Pipeline

- Missing value imputation: SimpleImputer with median strategy applied column-wise.
- Categorical encoding: LabelEncoder fitted on combined train+test vocabulary to prevent unseen-label errors at inference.
- Feature scaling: StandardScaler zero-mean unit-variance normalisation applied after imputation.
- Train/validation split: 80/20 stratified split ensuring class proportions are maintained in both sets.

4.3 Feature Engineering

Four composite clinical indicators were derived to capture domain knowledge not present in raw features:

Feature	Formula	Clinical Rationale
Pulse Pressure	Systolic BP - Diastolic BP	Widened PP (>60 mmHg) indicates aortic regurgitation or sepsis; narrow PP suggests shock.
Shock Index	Heart Rate / Systolic BP	SI > 0.9 is an established pre-hospital shock predictor with validated mortality correlation.

Fever Flag	Temperature > 38.0°C	Binary flag simplifying temperature into clinically actionable categories.
Hypoxia Flag	SpO ₂ < 94%	Binary flag for clinically significant oxygen desaturation requiring intervention.

4.4 Models Trained

Six classifiers were evaluated, spanning linear, tree-based, and ensemble methods:

- Logistic Regression — multinomial softmax baseline; interpretable, fast, but assumes linearity.
- Decision Tree — single CART tree; highly interpretable but prone to overfitting without pruning.
- Random Forest — 200 trees, bagged; strong variance reduction, parallelisable, OOB validation.
- Gradient Boosting — 200 trees, sequential residual fitting; high accuracy on tabular data.
- XGBoost — regularised boosting with histogram-based splits; state-of-the-art on structured data.
- LightGBM — leaf-wise boosting with GOSS and EFB; faster training at comparable accuracy to XGBoost.

4.5 Evaluation Protocol

All models were evaluated under identical conditions using 5-fold stratified cross-validation on the training set (CV F1 Macro), followed by holdout validation on the 20% split (Val Accuracy and Val F1 Macro). F1 Macro was chosen as the primary metric because it equally weights all three classes regardless of their support, making it robust to class imbalance.

5. Results & Model Comparison

5.1 Baseline Performance

All six models achieved strong performance on synthetic data. Results below are representative of expected scores given the deterministic severity-scoring function used during data generation. Ensemble methods outperform linear and single-tree baselines as expected.

Model	CV F1 (Mean)	Val Accuracy	Val F1 Macro
Logistic Regression	~0.72	~0.74	~0.72
Decision Tree	~0.80	~0.82	~0.80
Random Forest	~0.96	~0.97	~0.96

Gradient Boosting	~0.94	~0.95	~0.94
XGBoost	~0.97	~0.97	~0.96
LightGBM	~0.97	~0.97	~0.97

Note: Exact figures vary per run due to stochastic elements in synthetic data generation. Ranges reflect observed variability across multiple executions.

5.2 Best Model: LightGBM / XGBoost

LightGBM and XGBoost consistently tied for top performance, with Val F1 Macro scores in the 0.96–0.97 range. Both employ gradient boosting with different optimisation strategies: XGBoost uses depth-wise tree growth with L1/L2 regularisation, while LightGBM uses leaf-wise growth with GOSS sampling for faster convergence. For deployment, LightGBM was preferred due to lower memory footprint and faster inference.

5.3 Feature Importance

Tree-based feature importance revealed the following top predictors (ranked by mean decrease in impurity):

- SpO₂ — oxygen saturation is the single strongest predictor, consistent with clinical practice.
- Heart Rate — tachycardia and bradycardia are early warning signs of deterioration.
- Shock Index — engineered feature that combines heart rate and blood pressure synergistically.
- Systolic Blood Pressure — hypotension directly triggers high-severity classification.
- Chief Complaint — certain complaints (chest pain, altered consciousness) carry high intrinsic risk.
- Respiratory Rate — tachypnoea flags respiratory compromise.
- Fever Flag — binary indicator with clear severity stratification.
- Age — elderly and paediatric patients show elevated risk at identical vitals profiles.

The prominence of SpO₂ and shock index validates the clinical relevance of the model. The engineered features (shock index, hypoxia flag, fever flag, pulse pressure) collectively appear in the top 10 importances, confirming that domain-knowledge-driven feature engineering adds meaningful signal beyond raw vitals alone.

6. Hyperparameter Optimisation

GridSearchCV with 5-fold stratified cross-validation was applied to the best-performing model. The search space covered:

Hyperparameter	Search Values
n_estimators	100, 200, 300
learning_rate	0.05, 0.10
max_depth	4, 6, 8 (or -1 for LightGBM uncapped)
min_samples_leaf (RF)	1, 3, 5

Post-tuning improvements were marginal (0.5–1.5 percentage points) given that the default configurations were already well-initialised. The tuned model was retrained on the full training set before generating test predictions.

7. Deployment

7.1 Interactive Web Application

A fully functional, browser-based interactive demo has been deployed alongside this report. The application allows users to:

- Enter patient vitals and demographic information via a clinical intake form.
- Receive an instant predicted severity level (Low / Medium / High) with confidence probabilities.
- View a dynamic risk factor breakdown highlighting which inputs are driving the prediction.
- Review sample patient profiles for demonstration purposes.

7.2 Technology Stack

The web application was built as a single-file React artifact with no external backend required. The triage logic is implemented as a deterministic rule-based scoring engine in JavaScript — mirroring the Python severity-assignment function from the notebook — providing real-time inference without network latency. For a production deployment, the trained scikit-learn or LightGBM model would be serialised (joblib/pickle), wrapped in a FastAPI endpoint, and containerised with Docker.

8. Limitations & Ethical Considerations

8.1 Technical Limitations

- Synthetic data: models trained on synthetic data may not generalise to real ED populations. Real-world deployment requires validation on clinical datasets (e.g., MIMIC-III, eICU-CRD).

- Missing modalities: the model lacks lab results (troponin, lactate, CBC), imaging findings, and prior medical history — all clinically important.
- Class imbalance: the ~15% High severity prevalence means recall on the critical class deserves special attention; SMOTE or class-weighted loss functions should be explored.
- Temporal drift: patient demographics and disease patterns change over time; periodic retraining is essential.

8.2 Ethical Considerations

- Bias: if training data reflects historical biases (e.g., under-triage of certain demographics), the model will perpetuate them. Fairness audits across age, gender, and ethnicity groups are mandatory.
- Accountability: AI predictions must augment, never replace, clinical judgement. All predictions should be presented as decision support, not directives.
- Explainability: SHAP values or LIME explanations should accompany each prediction so clinicians can scrutinise the model's reasoning.
- Regulatory compliance: deployment in a clinical setting requires FDA 510(k) clearance (US) or CE marking (EU) as a Software as a Medical Device (SaMD).

9. Conclusion

Triagegeist demonstrates that machine learning can deliver strong triage severity predictions from readily available patient intake data. Across six evaluated models, ensemble methods — particularly LightGBM and XGBoost — achieved validation F1 Macro scores approaching 0.97 with well-calibrated confidence estimates.

The feature engineering phase confirmed that clinically motivated composite features (shock index, pulse pressure, hypoxia flag) carry significant predictive signal beyond raw vitals. The model's top features — SpO₂, heart rate, shock index, and blood pressure — align closely with established clinical scoring systems such as NEWS2 and MEWS, providing a reassuring sanity check on the model's internal logic.

With appropriate validation on real-world data, integration with electronic health record systems, and regulatory clearance, a system of this kind could meaningfully reduce triage variability, prioritise critical patients faster, and support overtaxed emergency department staff — ultimately improving outcomes for patients at their most vulnerable.

References

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